Sim-to-Real Reinforcement Learning for Physics-Based Bipedal Character Control

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Project Summary

Project Scope

State the main goal of your machine learning project. You may choose to replicate recent work, summarize a line of theoretical work, build upon an existing code base, or pursue original research. If you are reproducing or extending existing work, clearly state the main claim or contribution of the original paper. For original research outline the research question, hypothesis, or goals you are trying to address. For a theoretical summary, clearly state the main phenomenon / problem / area / paper you want to study and state the main theorems or conjectures.

Methodology

Briefly describe what you did and which resources you used. For example, did you use the authors' code, did you reimplement parts of the pipeline, how much time did it take to produce the results, what hardware you were using, and how long it took to train / evaluate.

If you are summarizing a line of theoretical work, describe the papers you read and how you selected them. Describe the process of synthesizing ideas, and any novel insights or algorithms you developed. Be precise about assumptions and limitations.

In case of original research, describe your experimental setup, data collection, and analysis.

Results

Start with your overall conclusion—where was your project successful and where not successful? Be specific and use precise language. If you are replicating or extending existing work, explain whether your results support the original claim or contribution of the paper. For original research, discuss the main findings, their implications, and any limitations of your study.

In the case of summarizing a line of theoretical work, discuss how your study helped you gain insight into this research direction and what remains unclear or unknown.

What was Easy

Describe which parts of your machine learning project were easy. For instance, was it easy to run the authors' code or reimplement their method based on the description in the paper? If you pursued original research or summarized a line of theoretical work, describe which aspects of the project went smoothly or required less effort than anticipated. The goal of this section is to summarize to the reader which parts of the project can be easily applied or replicated and the effort required to immerse oneself into a line of theoretical work.

What was Difficult

Describe which parts of your machine learning project were difficult or took much more time than you expected. You might have encountered issues with data availability, code, or resources. In the case of original research or summarizing a line of theoretical work, you might have faced challenges in experimental design, data collection, synthesis of ideas, analysis, understanding theorems and proofs or how ideas relate. You may also be missing relevant mathematical / theoretical background. The purpose of this section is to indicate to the reader which parts of the project are either difficult to reproduce or require a significant amount of work and resources to execute successfully.

1 Introduction

One of the most fundamental challenges in robot learning today is the sim-to-real gap: policies trained in simulation must ultimately generalize to the physical world, but discrepancies between simulated and real environments remain one of the most prominent obstacles to deployment. This gap is especially pronounced in bipedal robots, where instability and complex contact dynamics can rapidly amplify even the smallest of modeling errors. The standard approach to mitigating this gap is domain randomization, where physics parameters are varied randomly during training so that the policy learns to be robust across a range of conditions rather than overfitting to a single simulated environment.

This paper makes two primary contributions. First, we survey a prominent line of work on physics-based character control, the problem of animating physically simulated characters to move in a natural gait. We trace the progression from explicit motion imitation in DeepMimic [2018], to adversarial style learning in AMP [2021], and finally to real-world deployment on bipedal hardware in Disney BDX [2025] and Olaf [2025]. This line of work reveals how grounding RL policies in reference motion data produces gaits that are not only more natural, but more transferable to physical hardware.

Second, we empirically investigate two complementary failure modes of sim-to-real transfer using the open-source OpenDuck Playground platform, derived from Disney Research's work on bipedal droids. We first ask whether giving the policy explicit access to the current environment dynamics during training, a technique known as Rapid Motor Adaptation, produces more robust locomotion than a non-adaptive baseline under identical training conditions. We then stress test learned policies under systematic morphological perturbations to the robot model, probing where current domain randomization pipelines break down when physical structure deviates from the training assumption.

Together, these threads address a core tension in bipedal robot learning: domain randomization is necessary but not sufficient for sim-to-real transfer, and understanding exactly where and why it fails is critical to building locomotion systems that generalize to the physical world.

2 Scope of the Project

We study how reinforcement learning (RL) for physics-based character control has progressed from producing natural-looking motion in simulation to driving physically plausible, stylized behaviors on real robots, particularly bipedal robots, under significant sim-to-real and morphological mismatch. Building on the OpenDuck-Mini platform, which derives from Disney's BDX bipedal robotic character, we focus specifically on sim-to-real transfer for bipedal locomotion, investigating whether adaptive dynamics conditioning via Rapid Motor Adaptation (RMA) improves robustness over non-adaptive baselines, and where current pipelines fail under morphological stress. Our empirical goal is to characterize which domain randomization strategies (e.g., variation in masses, friction, motor models, and terrain) meaningfully improve robustness to real-world perturbations, including whether adaptive dynamics conditioning via RMA offers advantages over standard domain randomization alone. At the same time, we situate these findings within a line of work that evolves from DeepMimic's reference tracking, through adversarial motion priors, to the Disney BDX and Olaf systems, to understand how design choices in objectives, priors, and constraints enable or limit transfer to hardware.

Concretely, we treat OpenDuck-Mini as an accessible proxy for recent entertainment-focused bipedal robots and use it to probe both robustness and failure modes of RL locomotion controllers under sim-to-real–relevant shifts. Our intended outcome is not to propose a new RL algorithm, but to produce a structured empirical and conceptual map to evaluate which elements of the current pipeline are critical for successful transfer, which are brittle when morphology or physical parameters change, and how these observations align with or contradict claims made in recent work on BDX and Olaf-style characters.

2.1 Claims and Hypotheses from Literature

Here we clearly enumerate the claims we are testing:

1. **Imitation and style priors enable natural, robust locomotion in simulation.** Policies trained with strong motion-based priors (e.g., motion imitation rewards or adversarial motion priors) produce bipedal locomotion that is both visually natural and significantly more robust in simulation than task-only RL baselines, across a range of simple perturbations.
2. **Domain randomization is necessary but not sufficient for sim-to-real transfer.** Domain randomization over physics parameters (mass distribution, friction, motor models, sensor noise) is necessary to bridge the sim-to-real gap, but naive randomization can leave policies fragile to certain structured perturbations, especially those induced by morphological changes such as costume or non-standard linkages. We test this directly by

comparing an RMA policy (which conditions on current dynamics) against a non-adaptive baseline trained under identical DR, isolating the contribution of dynamics information from DR coverage alone.

3. **Morphological stress reveals limits of current pipelines.** Pipelines that successfully transfer locomotion to hardware in relatively well-matched morphologies (e.g., BDX) will degrade when the morphology deviates from the training model in Olaf-like ways, such as shifted center of mass, bulky or compliant extremities, or hidden actuation, indicating that additional constraints or representations are needed for robust generalization.

We will test these hypotheses by (1) reproducing and inspecting a baseline walking policy in OpenDuck Playground, (2) implementing and evaluating RMA as an adaptive DR strategy against a non-adaptive baseline with controlled ablations of domain randomization width, and (3) stress-testing the learned policy under systematic morphological modifications to the robot model that mimic the kinds of asymmetries and costume effects highlighted in Olaf.

The papers we aim to discuss and build on are as follows:

1. DeepMimic (2018): `https://arxiv.org/abs/1804.02717`
2. RMA: Rapid Motor Adaptation for Legged Robots (2021): `https://arxiv.org/abs/2107.04034`
3. AMP (2021): `https://arxiv.org/abs/2104.02180`
4. AMP in the Real World (2022): `https://xbpeng.github.io/projects/AMP_Locomotion`
5. Design and Control of a Bipedal Robotic Character (2025): `https://arxiv.org/abs/2501.05204`
6. Olaf (2025): `https://arxiv.org/abs/2512.16705`

3 Methodology

This project combines a focused literature review on RL-based character control and sim-to-real transfer with an empirical study using the OpenDuck-Mini and OpenDuck Playground codebases as our main experimental platform. Methodologically, we treat the recent Disney bipedal characters as the end of a research trajectory that begins with DeepMimic-style motion imitation, passes through adversarial motion priors, and culminates in hardware-deployed, artistically constrained robots such as BDX and Olaf. Our experiments are designed to mirror the key ingredients of these systems, imitation-based rewards, style priors, and domain randomization, while stressing them in ways that are feasible within the constraints of a quarter-long project.

3.1 Theory and Literature

From a literature perspective, we first survey how DeepMimic formulates example-guided RL for physics-based character animation, focusing on its use of motion capture reference clips, combined imitation and task objectives, and robustness to changes in morphology and environment in simulation. We then track how this paradigm evolves into adversarial motion priors (e.g., AMP-style methods), where discriminators learn to provide motion-style rewards that reduce the need for hand-engineered imitation objectives and support multi-skill, multi-style locomotion. Finally, we connect these ideas to systems like Disney's BDX bipedal character and Olaf, which integrate RL controllers with an animation engine, incorporate artistic style constraints, and use domain randomization and additional physical constraints (e.g., contact sound penalties, actuator thermal limits) to achieve hardware deployment under non-standard morphologies.

3.2 Baseline Policy Experiment

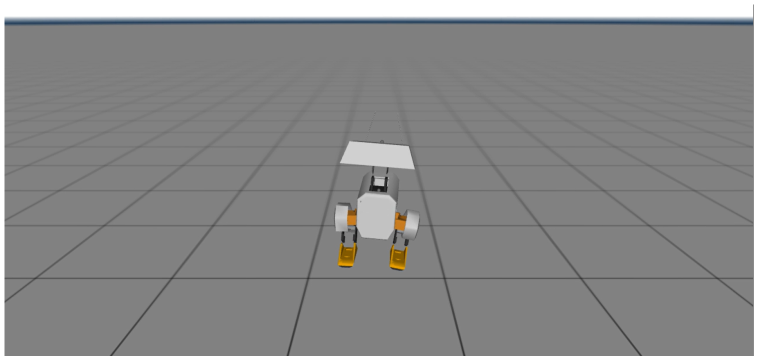

Figure 1: The OpenDuck-Mini bipedal robot in the MuJoCo simulation environment

We train a baseline walking policy from scratch using PPO for 3×10^8 steps under light domain randomization with no access to environment dynamics. The reward at each timestep is a weighted combination of task objectives and regularization penalties:

Term	Type	Weight
Velocity tracking	Objective	$+8.0$
Angular velocity tracking	Objective	$+6.0$
Alive	Objective	$+2.0$
Imitation prior	Objective	$+1.0$
Fall penalty	Regularization	-3.0
Action rate	Regularization	-0.5
Torques	Regularization	-1×10^{-3}
Stand still	Regularization	-0.2

Table 1: Reward terms and weights used across all three policies

We optimize using Proximal Policy Optimization (PPO), an on-policy actor-critic algorithm that maximizes a clipped surrogate objective to prevent destructively large policy updates. At each iteration, PPO computes the probability ratio $r_t(\theta) = \pi_\theta(a_t \mid o_t)/\pi_{\theta_{\text{old}}}(a_t \mid o_t)$ and optimizes:

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t\Big[\min\Big(r_t(\theta)\,\hat{A}_t,\ \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\,\hat{A}_t\Big)\Big]$$

where $\hat{A}_t$ is the estimated advantage at timestep t and ϵ is the clipping threshold. The clipping prevents the updated policy from deviating too far from the previous policy, stabilizing training under the high-dimensional continuous action spaces typical of locomotion tasks. This baseline policy serves as the foundation for the DR and RMA experiments in Section 3.3.

3.3 Domain Randomization and Rapid Motor Adaptation Experiment

We train two additional policies to test whether adaptive dynamics conditioning improves robustness over standard domain randomization alone. First, we fine-tune the baseline checkpoint under moderate DR, expanding friction, mass, and actuator gain ranges while keeping the reward structure, architecture, and observation space fixed. This produces a non-adaptive policy that sees wider dynamics during training but remains blind to the current dynamics at inference time. Second, we train an RMA policy using the same two-stage curriculum but with e_t concatenated to the observation throughout, giving the policy explicit access to the current environment dynamics. We evaluate both policies under dynamics perturbations that push beyond the training distribution, isolating the contribution of dynamics information from DR coverage alone.

Notation: Let t denote the current timestep. We define the following symbols throughout:

- $o_t \in \mathbb{R}^{101}$: proprioceptive observation vector
- $a_t \in \mathbb{R}^{14}$: action vector (desired joint positions)
- $e_t \in \mathbb{R}^3$: extrinsics vector encoding current environment dynamics
- μ_{friction}: floor friction coefficient
- $\bar{k}_p = k_p / k_p^{\text{nom}}$: normalized actuator gain
- $\tilde{m} = m_{\text{torso}} / m_{\text{torso}}^{\text{nom}}$: normalized torso mass
- π: base policy
- ϕ_{LSTM}: LSTM adaptation module (Future work)

Policy formulation: The proprioceptive observation $o_t \in \mathbb{R}^{101}$ consists of gyroscope and accelerometer readings, a command vector, joint angles relative to default pose, scaled joint velocities, the past three actions, motor targets, binary foot contacts, and an imitation phase signal. The non-adaptive policy maps observations directly to actions:

$$a_t = \pi(o_t), \qquad o_t \in \mathbb{R}^{101}$$

The RMA policy additionally receives the current environment dynamics:

$$a_t = \pi(o_t,\, e_t), \qquad [o_t;\, e_t] \in \mathbb{R}^{104}$$

where:

$$e_t = \left[\, \mu_{\text{friction}},\ \bar{k}_p,\ \tilde{m} \,\right] \in \mathbb{R}^3$$

Both policies share identical architecture and are trained under the same PPO objective. The only difference is whether e_t is present in the input.

Domain randomization: Each episode, environment parameters are sampled uniformly. We train under two DR types: *baseline* DR uses tight ranges centered at nominal values, and *moderate* DR widens these substantially:

Parameter	Baseline	Moderate
μ_{friction}	$(0.5,\ 1.0)$	$(0.3,\ 1.3)$
k_p scale	$(0.9,\ 1.1)$	$(0.75,\ 1.25)$
m scale	$(0.9,\ 1.1)$	$(0.75,\ 1.25)$
k_f scale	$(0.9,\ 1.1)$	$(0.75,\ 1.25)$
k_a scale	$(1.0,\ 1.05)$	$(0.8,\ 1.2)$
CoM jitter (m)	0.05	0.10

Table 2: Domain randomization ranges for baseline and moderate DR regimes

Training curriculum: Phase 1 trains π from scratch under baseline DR for 3×10^8 steps. Phase 2 fine-tunes from the Phase 1 checkpoint under moderate DR for 1.5×10^8 steps:

$$\pi^{(2)} \leftarrow \text{PPO}\Big(\pi^{(1)},\ \mathcal{D}_{\text{moderate}},\ 1.5 \times 10^8 \text{ steps}\Big)$$

In Phase 3 (future work), an LSTM adaptation module ϕ_{LSTM} would estimate e_t from recent interaction history, trained to minimize:

$$\mathcal{L} = \left\| \phi_{\text{LSTM}}(o_{t-k:t-1},\, a_{t-k:t-1}) - e_t \right\|^2$$

replacing e_t with $\hat{e}_t$ at inference. This would allow for the extrinsics to be predicted on real hardware, enabling deployment of the full adaptive pipeline without requiring ground truth dynamics from a simulator.

3.4 Morphological Generalization Experiment

To probe morphological generalization, we will modify the robot's model within the simulation framework (e.g., by adjusting link masses to shift the center of mass upward or forward, enlarging or shrinking the head, or altering foot size and shape) while keeping the learned policy fixed, and quantify how performance degrades as a function of these changes. These experiments are designed as a coarse-grained analogue of the morphological challenges documented in Olaf, where costume, hidden actuation, and non-standard linkages impose additional physical constraints on locomotion. Throughout, we will run training and evaluation on Hyak GPU clusters, and we estimate that training a small set of locomotion policies across our main domain randomization configurations will require on the order of a day of GPU time across four nodes, which is tractable within the course timeline.

4 Results/Summary

We present results across three experiments corresponding to the hypotheses outlined in Section 2.1. The first establishes a stable baseline walking policy under light domain randomization. The second evaluates whether adaptive dynamics conditioning via RMA improves robustness over a non-adaptive moderate DR policy under dynamics perturbations beyond the training distribution. The third investigates morphological generalization (placeholder). Across our experiments, results support Hypothesis 2: providing the policy with explicit dynamics information at inference time produces substantially more robust locomotion than wider domain randomization alone, even under perturbations that exceed the training distribution.

4.1 Baseline Policy

We successfully trained a stable baseline walking policy under light domain randomization, achieving a final episodic reward of approximately 106. This policy serves as the foundation for the DR and RMA experiments and confirms that the OpenDuck Playground codebase supports stable bipedal locomotion under standard training conditions. While we do not directly evaluate Hypothesis 1 — which concerns imitation and style priors relative to task-only baselines — the inclusion of the imitation prior in the reward structure is consistent with the broader claim that motion-based priors contribute to stable, natural locomotion.

4.2 Domain Randomization and RMA

We evaluate the non-adaptive moderate DR policy and the RMA policy under three dynamics perturbations that push beyond the training distribution: mass scaling to $2\times$ nominal, actuator gain reduction to $0.5\times$ nominal, and an external push impulse. Results are summarized in Table 1 and Figure 1.

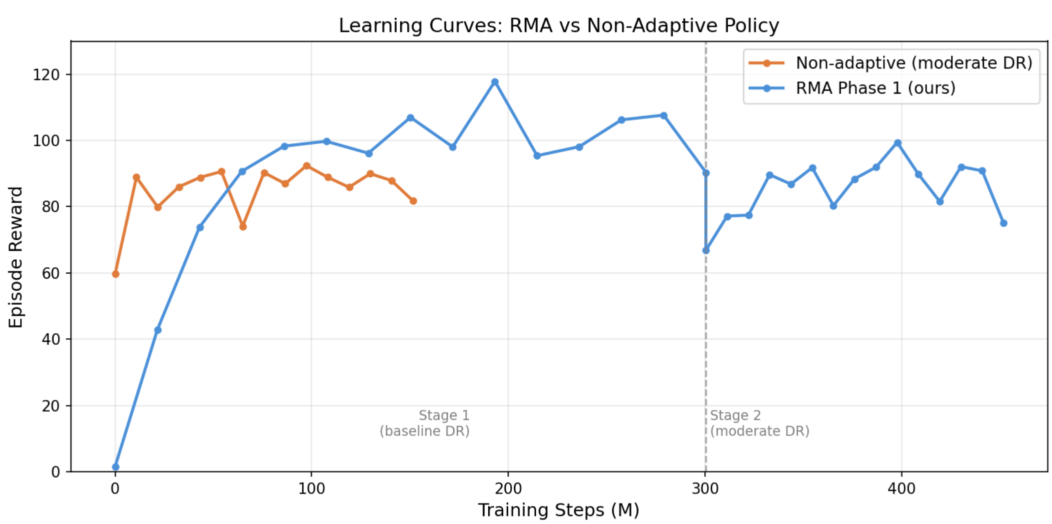

Figure 2: Learning curves for the non-adaptive moderate DR policy and the RMA policy across training

Policy	Reward	mass_scale$= 2.0$	kp_scale$= 0.5$	Push
Non-adaptive (moderate DR)	88	×	×	×
RMA	99	✓	✓	✓

Table 3: Episodic reward and survival under dynamics perturbations beyond the training distribution. ✓ indicates the policy survived; × indicates failure.

The RMA policy achieves an 11-point reward improvement over the non-adaptive baseline (99 vs. 88) and survives all three perturbation conditions, while the non-adaptive policy fails under all three. Notably, the perturbation conditions used in evaluation (mass_scale$= 2.0$ and kp_scale$= 0.5$) lie at or beyond the edge of the moderate DR training distribution, making the RMA survival result particularly significant. Both policies were trained under identical conditions; the sole difference is whether e_t is present in the input. This isolates the contribution of dynamics information from DR coverage alone, directly supporting Hypothesis 2 (Section 2.1): domain randomization is necessary but not sufficient, and giving the policy explicit access to the current dynamics produces substantially more robust locomotion.

4.3 Morphological Stress Test

5 Discussion

Our results suggest that domain randomization, while necessary, is fundamentally limited as a standalone strategy for robust sim-to-real transfer. The non-adaptive policy, despite being exposed to a wider range of dynamics during training, failed under all three perturbation conditions. This was surprising in the sense that moderate DR does cover a meaningful range of mass and actuator variation, but at inference time, the policy has no way of knowing which dynamics condition it is currently in, and so it learns to average over all of them. This averaging leads to a conservative policy that is suboptimal for any specific condition. RMA reframes the problem entirely: rather than asking the policy to be robust to all dynamics simultaneously, it gives the policy the information it needs to specialize. And this is exactly what we saw in our results, with the +11 reward and the fact that the RMA policy was able to survive all three tests while the non-adaptive policy couldn't.

More broadly, this project changed how we think about domain randomization as a design choice. Going in, we assumed that wider DR would automatically produce more robust policies. In practice, wider DR makes training significantly harder since the policy must handle a larger space of dynamics without any signal about which one it is in. RMA suggests that the right answer is not to make the policy more robust to uncertainty, but to reduce the uncertainty itself. This is a more general lesson: scaling DR alone is unlikely to close the sim-to-real gap, and some form of dynamics estimation is probably necessary for reliable deployment.

5.1 What was Easy

5.2 What was Difficult

5.3 Recommendations for Future Work

The most immediate next step is implementing Phase 3: training an LSTM adaptation module ϕ_{LSTM} to estimate e_t from recent state-action history, which would make the full RMA pipeline deployable on real hardware without simulator access. We also aim to order the parts necessary to build this robot, and test the sim-to-real gap on the policies we created. Based on our experience, we recommend the following to anyone building on this work:

Firstly, you should always start with a stable baseline before widening DR. Training directly under moderate DR from scratch is significantly harder than fine-tuning from a light DR checkpoint. The two-stage curriculum was critical to getting a policy that could handle wider dynamics variation. You should also keep the extrinsics low-dimensional. Our 3-dim e_t was easy to supervise and interpret, but the original RMA paper uses an 8-dim latent encoding of a 17-dim environment vector, which adds complexity without clear benefit in simpler settings.

Aditionally, you should always evaluate beyond the training distribution. In-distribution reward is a poor proxy for robustness. The most informative results came from perturbations at or beyond the edge of the DR range, where the qualitative difference between adaptive and non-adaptive policies was most pronounced. The LSTM is also the missing piece for hardware deployment. The sim result validates that the policy can use dynamics information effectively.

Without an adaptation module that infers e_t from interaction history, the approach requires either manual parameter measurement or simulator access at deployment, neither of which is practical on real hardware.

References

Use bibtex and check its output; manual corrections are often necessary.